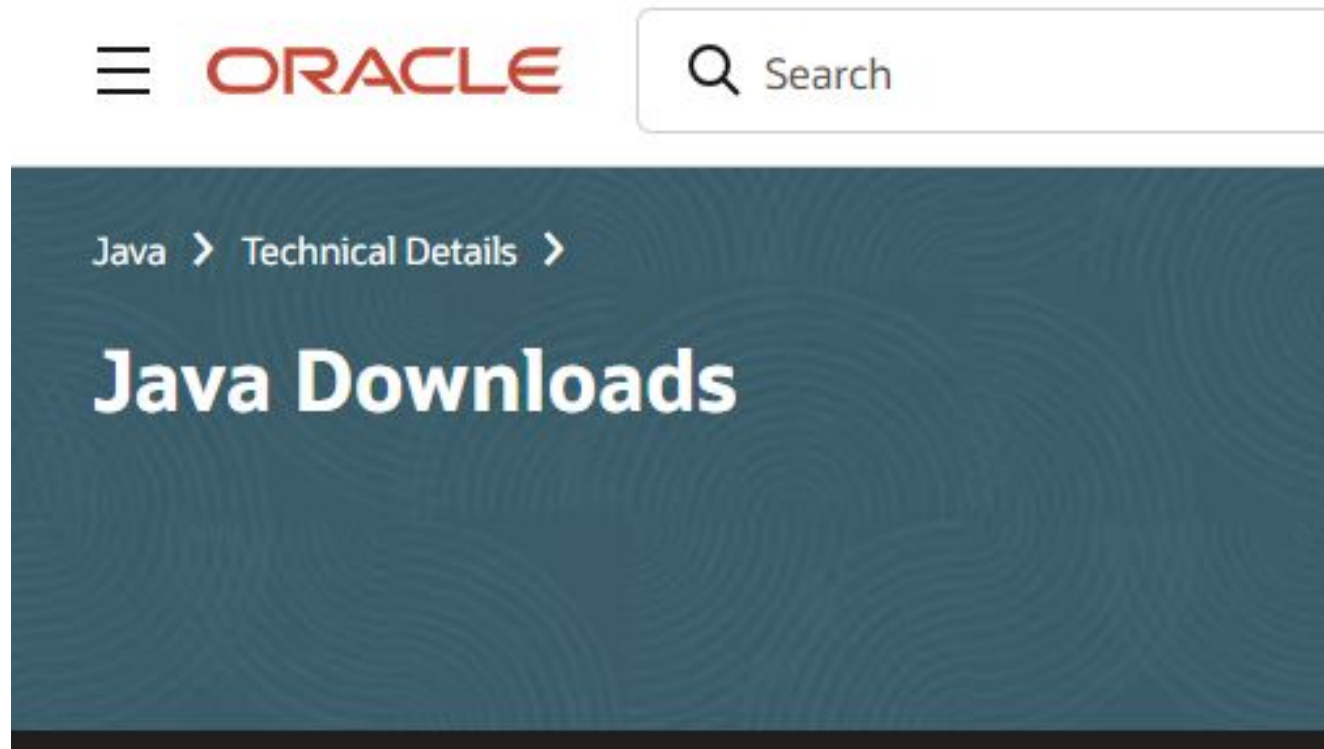


Implementing a CI/CD pipeline
with Jenkins automates your
software's build, test, and
deployment process

Step 1: Install Jenkins

- A. Install Java (JDK 17 or later).
- B. Download and install Jenkins.
- C. Start the Jenkins service.
- D. Open Jenkins in your browser:
- E. <http://localhost:8080>
- F. Unlock Jenkins using the initial administrator password.
- G. Install the suggested plugins.
- H. Create the first admin user.

Download java 21 and above



Download java msi installer

JDK 26 JDK 25 JDK 21

Java SE Development Kit 25.0.4 downloads

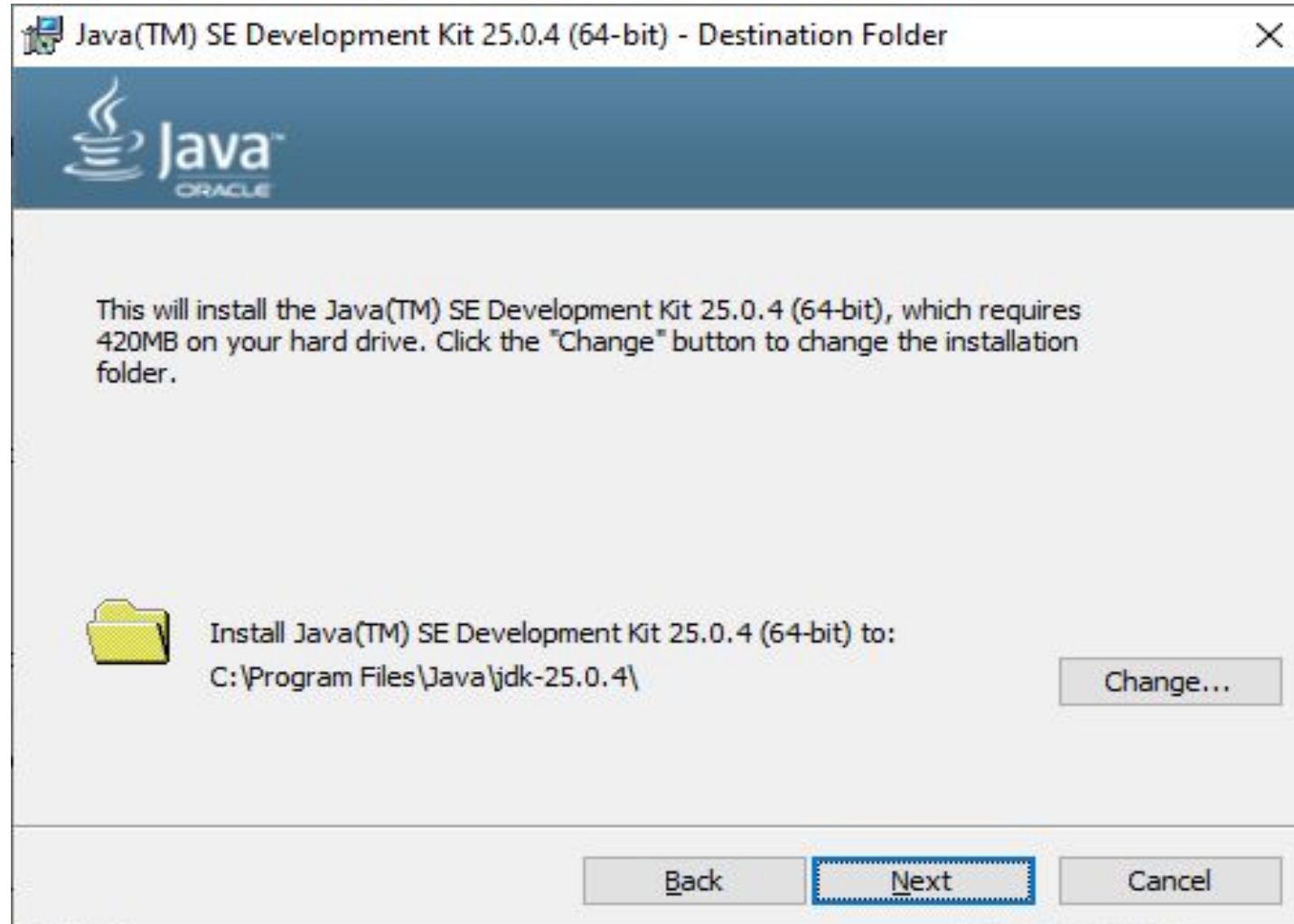
JDK 25 binaries are free to use in production and free to redistribute, at no cost, under the [Oracle No-Fee Terms and Conditions \(NFTC\)](#).

JDK 25 will receive updates under the NFTC, until September 2028, a year after the release of the next LTS. Subsequent JDK 25 updates will be licensed under the [Java SE License \(OTN\)](#) and production use beyond the [limited free grants](#) of the OTN license will [require a fee](#).

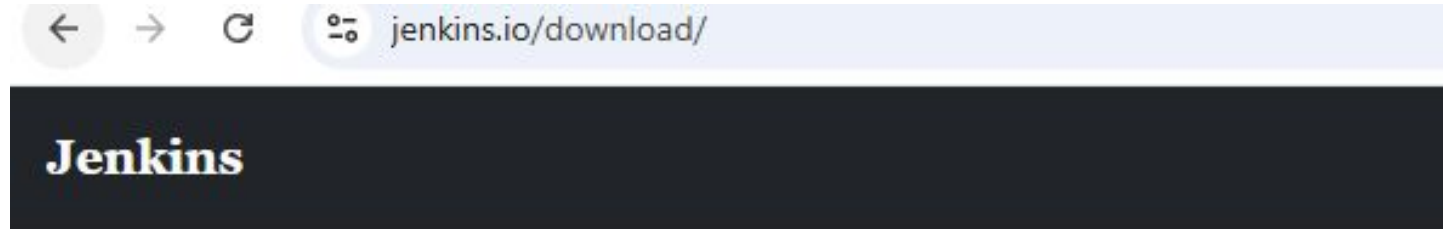
Linux macOS Windows

Product/file description	File size	Download
x64 Compressed Archive	205.81 MB	https://download.oracle.com/java/25/latest/jdk-25_windows-x64_bin.zip (sha256)
x64 Installer	184.58 MB	https://download.oracle.com/java/25/latest/jdk-25_windows-x64_bin.exe (sha256)
x64 MSI Installer	183.35 MB	https://download.oracle.com/java/25/latest/jdk-25_windows-x64_bin.msi (sha256)

Install java



Download and install Jenkins.



Download and deploy

The Jenkins project produces two release lines: Stable (LTS) and preferred over the other. See the links below for more informat

Download Jenkins 2.568.1 LTS for:



Generic Java package (.war)

SHA-256:

58f24f3965fbef7708629fbe158d51bf138ffd577cadbc86b46367e8ad0beb83



Docker



Kubernetes



Ubuntu/Debian



Red Hat Enterprise Linux and derivatives



Fedora



Windows



openSUSE

Download Jenkins 2.574 for:



Generic Java package (.war)

SHA-256:

e924a55dff7e96efed4844df55c6433b8dc03c7e639b45f8289e6df8b8



Docker



Ubuntu/Debian



Red Hat Enterprise Linux and derivatives



Fedora



Windows



openSUSE

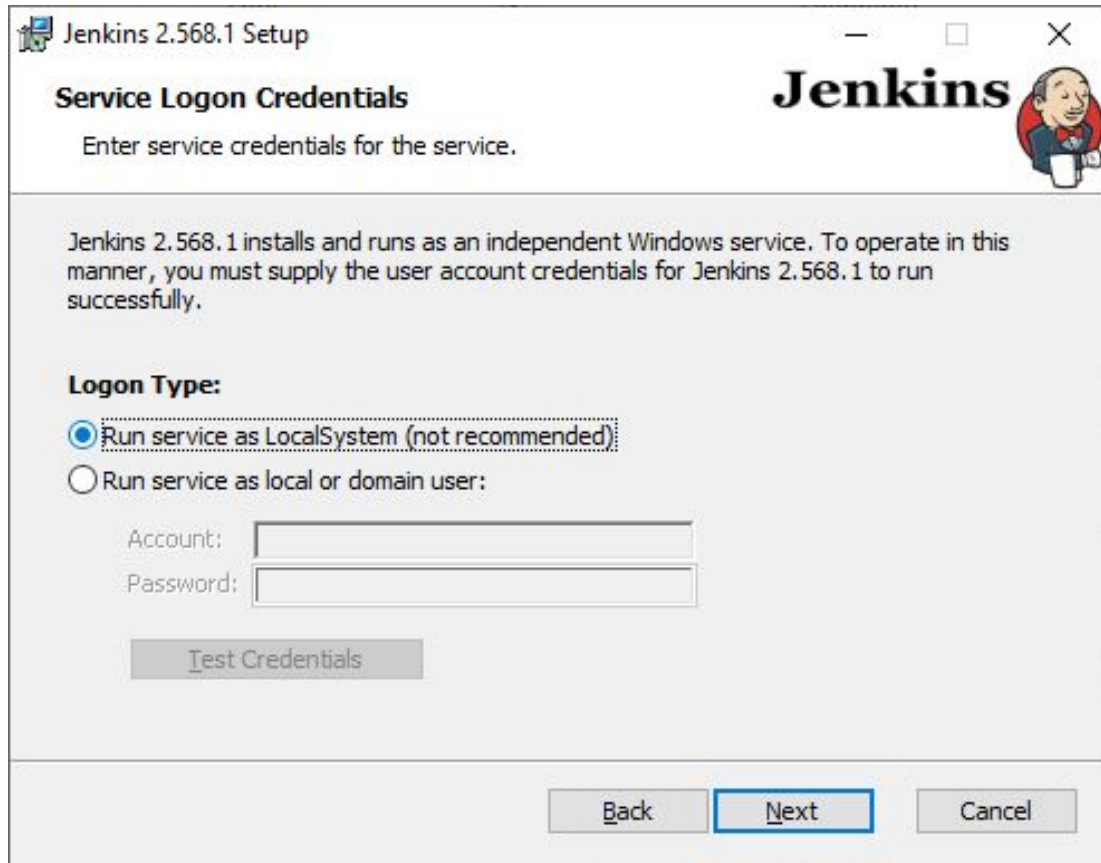


Arch Linux

Jenkins install



Select run service as local



The image shows a Windows installer window titled "Jenkins 2.568.1 Setup". The window has a standard Windows title bar with minimize, maximize, and close buttons. The main content area is titled "Service Logon Credentials" and includes the Jenkins logo (a cartoon character with a red bow tie) in the top right corner. Below the title, there is a text box for "Enter service credentials for the service." and a paragraph explaining that Jenkins 2.568.1 installs and runs as an independent Windows service, requiring user account credentials to operate successfully. Under the "Logon Type:" section, there are two radio button options: "Run service as LocalSystem (not recommended)" (which is selected) and "Run service as local or domain user:". Below these options are two text input fields labeled "Account:" and "Password:". A "Test Credentials" button is located below the password field. At the bottom of the window, there are three buttons: "Back", "Next" (which is highlighted with a blue border), and "Cancel".

Jenkins 2.568.1 Setup

Service Logon Credentials

Enter service credentials for the service.

Jenkins

Jenkins 2.568.1 installs and runs as an independent Windows service. To operate in this manner, you must supply the user account credentials for Jenkins 2.568.1 to run successfully.

Logon Type:

☒ Run service as LocalSystem (not recommended)

☐ Run service as local or domain user:

Account:

Password:

Test Credentials

Back Next Cancel

Test your port

Jenkins 2.568.1 Setup

Port Selection

Choose a port for the service.

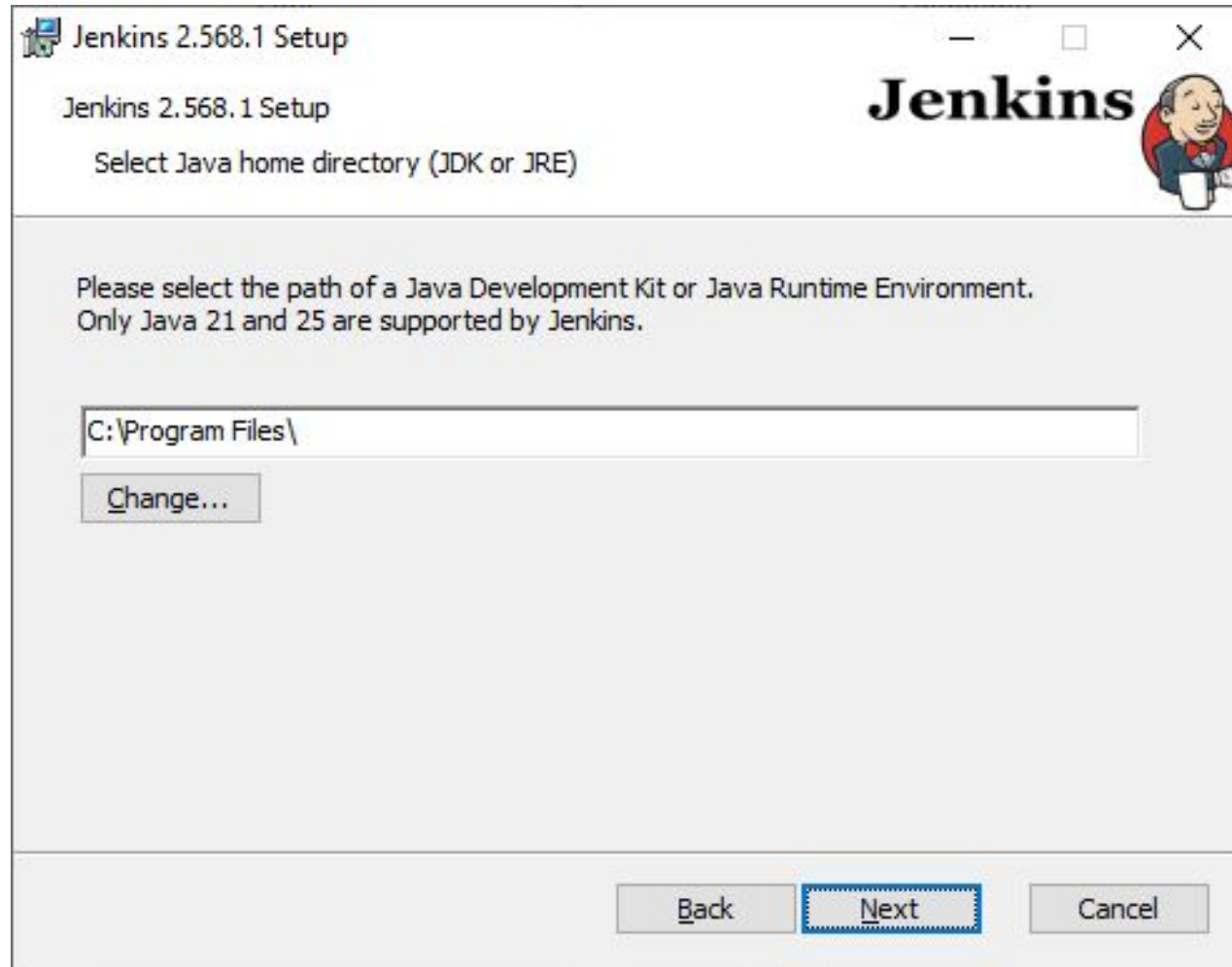
Please choose a port.

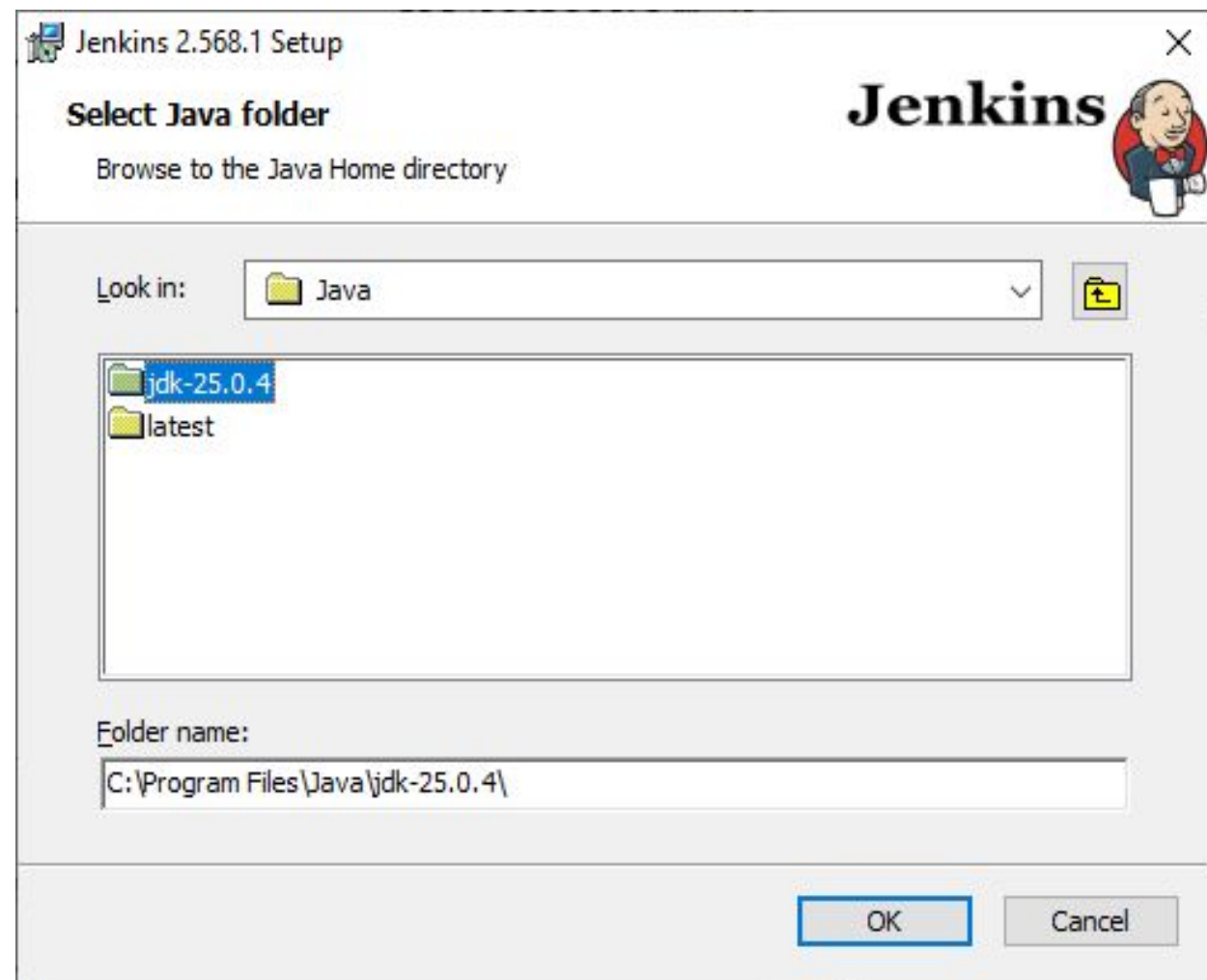
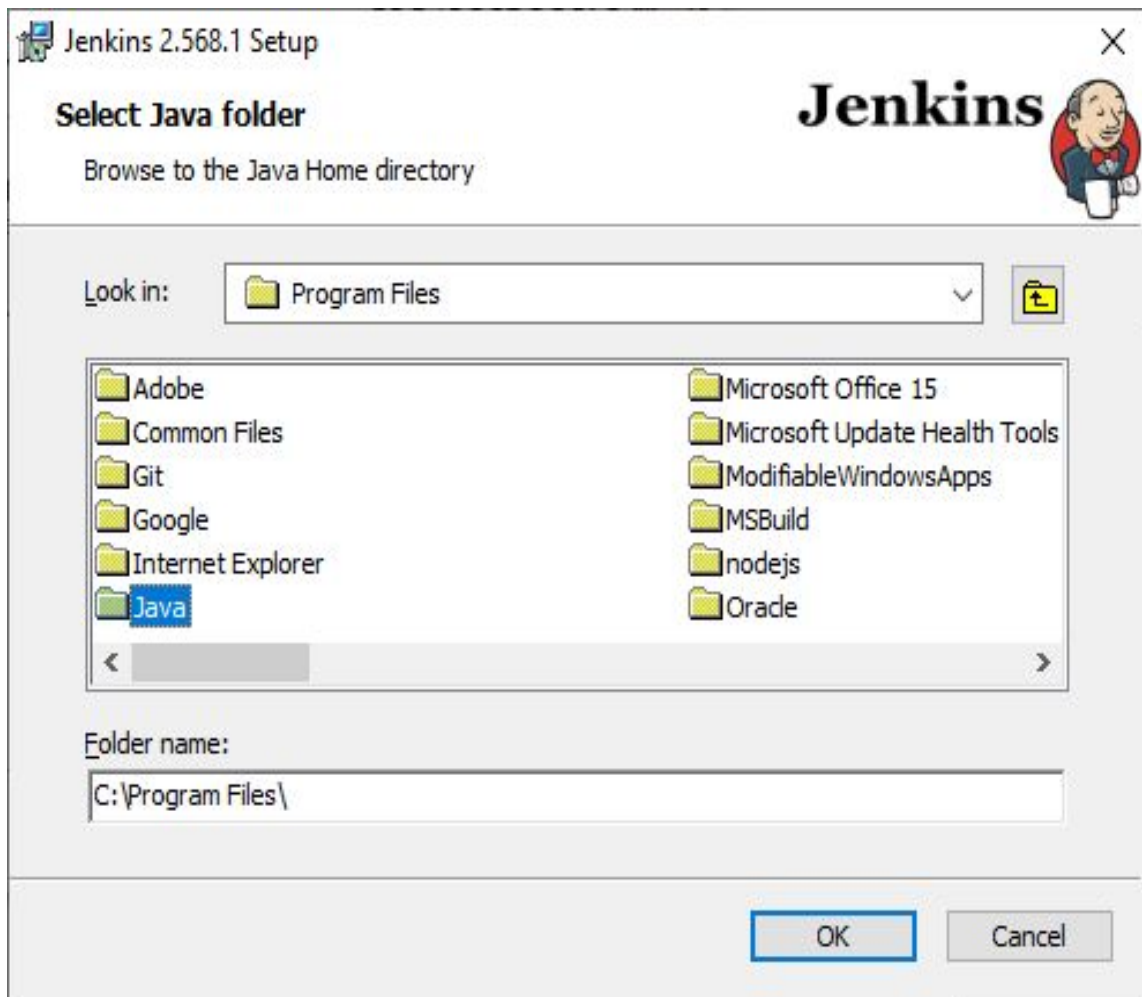
Port Number (1-65535):



It is recommended that you accept the selected default port.

Select java path folder





Jenkins 2.568.1 Setup

Completed the Jenkins 2.568.1 Setup
Wizard

Click the Finish button to exit the Setup Wizard.



Back

Finish

Cancel

Go to

- Type in the browser

Localhost:8080



localhost:8080/login?from=%2F

Getting Started

Unlock Jenkins

To ensure Jenkins is securely set up by the administrator, a password has been written to the log ([not sure where to find it?](#)) and this file on the server:

```
C:\ProgramData\Jenkins\.jenkins\secrets\initialAdminPassword
```

Please copy the password from either location and paste it below.

Administrator password

Copy this path

Getting Started

Unlock Jenkins

To ensure Jenkins is securely set up by the administrator, a password has been written to the log (not sure where to find it?) and this file on the server:

`C:\ProgramData\Jenkins\.jenkins\secrets\initialAdminPassword`

Please copy the password from either location and paste it below.

Administrator password

Continue

Paste in the browser you will get password



Paste and continue in the page

Getting Started

Unlock Jenkins

To ensure Jenkins is securely set up by the administrator, a password has been written to the log ([not sure where to find it?](#)) and this file on the server:

```
C:\ProgramData\Jenkins\.jenkins\secrets\initialAdminPassword
```

Please copy the password from either location and paste it below.

Administrator password

Continue

Customize Jenkins by install plugins

Getting Started



Customize Jenkins

Plugins extend Jenkins with additional features to support many different needs.

Install suggested plugins

Install plugins the Jenkins community finds most useful.

Select plugins to install

Select and install plugins most suitable for your needs.

Installing plugins

Getting Started

Getting Started

✓ Folders	✓ OWASP Markup Formatter	✓ Build Timeout	✓ Credentials Binding	** bouncycastle API
✓ Timestamper	⌚ Workspace Cleanup	⌚ Ant	⌚ Gradle	** Credentials
⌚ Pipeline	⌚ GitHub Branch Source	⌚ Pipeline: GitHub Groovy Libraries	⌚ Pipeline Graph View	** Plain Credentials
⌚ Git	⌚ SSH Build Agents	⌚ Matrix Authorization Strategy	⌚ LDAP	** Variant
⌚ Email Extension	⌚ Mailer	⌚ Dark Theme		** SSH Credentials
				Credentials Binding
				** SCM API
				** Pipeline: API
				Timestamper
				** Caffeine API
				** Script Security
				** Pipeline: Supporting APIs
				** Plugin Utilities API
				** Font Awesome API
				** Bootstrap 5 API
				** Jackson Annotations 2 API
				** SnakeYAML Engine API
				** Woodstox Core API
				** Jackson 3 API
				** JQuery3 API
				** ECharts API
				** Prism API
				** - required dependency

Create First Admin User

Username

Password

Confirm password

Full name

Email

Create First Admin User

Username

sivajenkin

Password

.....

Confirm password

.....

Full name

Sivakumar

Email

sivakumar.kaliyappan@rajalakshmi.edu.in

Instance Configuration

Jenkins URL:

The Jenkins URL is used to provide the root URL for absolute links to various Jenkins resources. That means this value is required for proper operation of many Jenkins features including email notifications, PR status updates, and the BUILD_URL environment variable provided to build steps.

The proposed default value shown is **not saved yet** and is generated from the current request, if possible. The best practice is to set this value to the URL that users are expected to use. This will avoid confusion when sharing or viewing links.

Jenkins is ready!

Your Jenkins setup is complete.

[Start using Jenkins](#)



Jenkins

+ New Item

📁 Build History



📝 Add description

Build Queue



No builds in the queue.

Build Executor Status

0/2



Welcome to Jenkins!

This page is where your Jenkins jobs will be displayed. To get started, you can set up distributed builds or start building a software project.

Start building your software project

Create a job



Set up a distributed build

Set up an agent



Configure a cloud



Learn more about distributed builds



Step 2: Install Required Plugins

- **Manage Jenkins → Plugins**
 - Install plugins such as:
 - Git Plugin
 - Pipeline Plugin
 - Docker Plugin (optional)
 - Maven Integration Plugin
 - NodeJS Plugin (if using Node.js)
 - Blue Ocean (optional)
- Restart Jenkins after installation.

Step 3: Configure Global Tools

- Go to:
 - **Manage Jenkins → Tools**
 - Configure:
 - Git
 - JDK
 - Maven
 - NodeJS (if needed)
 - Docker (if deploying containers)



New Item

Enter an item name

mynewite

Select an item type



Pipeline

Build, test, and deploy using pipelines. Supports stages, parallel work, and



Freestyle project

Classic, general-purpose job type that checks out from up to one SCM, ex followed by post-build steps like archiving artifacts and sending email not



Multi-configuration project

Suitable for projects that need a large number of different configurations, environments, platform-specific builds, etc.



Folder

Creates a container that stores nested items in it. Useful for grouping things which is just a filter, a folder creates a separate namespace, so you can have

OK



Configure



General



Triggers



Pipeline



Advanced

General

Description

my first jenkins test

[Plain text](#) [Preview](#)☐Discard old builds ☐

Do not allow concurrent builds

☐

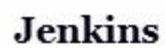
Do not allow the pipeline to resume if the controller restarts

☐

GitHub project

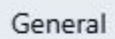
☐Pipeline speed/durability override ☐Preserve stashes from completed builds **Save**

Apply

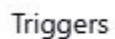


Jenkins / mynewite  / Configure

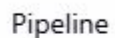
Configure



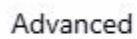
General



Triggers



Pipeline



Advanced

General

Description

my first jenkins test

Plain text [Preview](#)☐ Discard old builds ?☐ Do not allow concurrent builds

☐ Do not allow the pipeline to resume i

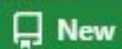
- ✓ GitHub project

Project url ?

Save

Apply

Top repositories



-  sivakumarkaliyappan-a11y/sivatest
-  sivakumarkaliyappan-a11y/sivatest2
-  sivakumarkaliyappan-a11y/local_testing
-  sivakumarkaliyappan-a11y/sivarepo
-  sivakumarkaliyappan-a11y/new_repository_test
-  sivakumarkaliyappan-a11y/remote_testing
-  sivakumarkaliyappan-a11y/tlfl6

Show more

Join GitHub Education!

GitHub Education opens doors to eager to drive innovation. Join us technology.

Free and discounted services for teachers and students.

[Join GitHub Education](#)

Copy repository url

The screenshot shows a GitHub repository page for 'sivakumarkaliyappan-a11y / sivatest'. The 'Code' button is clicked, opening a dropdown menu. The 'Local' tab is selected, and the 'Clone' option is chosen. The 'HTTPS' method is selected, and the URL 'https://github.com/sivakumarkaliyappan-a11y/sivatest' is displayed and circled in red. The 'SSH' and 'GitHub CLI' options are also visible. The 'Code' button is green and labeled '<> Code'. The repository is public and has 1 branch and 0 tags. The files listed are README.md, test.c, and test.exe, all from the first commit. The README content is '# sivatest'.

github.com/sivakumarkaliyappan-a11y/sivatest

sivakumarkaliyappan-a11y / sivatest

Code Issues Pull requests Agents Actions Projects Wiki Security and quality Insights Settings

sivatest Public

Pin Watch 0 Fork

main 1 Branch 0 Tags

Go to file T Add file <> Code

Local Codespaces

Clone ?

HTTPS SSH GitHub CLI

https://github.com/sivakumarkaliyappan-a11y/sivatest

Clone using the web URL.

Open in GitHub Copilot app

Open with GitHub Desktop

Download ZIP

sivakumarkaliyappan-a11y first commit

README.md first commit

test.c first commit

test.exe first commit

README

"# sivatest"

About

test

Read

Activ

0 sta

0 wa

0 for

Release:

No release

Create a new release

Package

Paste the url and save

General

Description

my first jenkins test

Plain text [Preview](#)

☐ Discard old builds [?](#)

☐ Do not allow concurrent builds

☐ Do not allow the pipeline to resume if the controller restarts

☒ GitHub project

Project url [?](#)

<https://github.com/sivakumarkaliyappan-a11y/sivatest.git>

Save

Apply

Paste pipeline scrip

Pipeline

Define your Pipeline using Groovy directly or pull it from source control.

Definition

Pipeline script

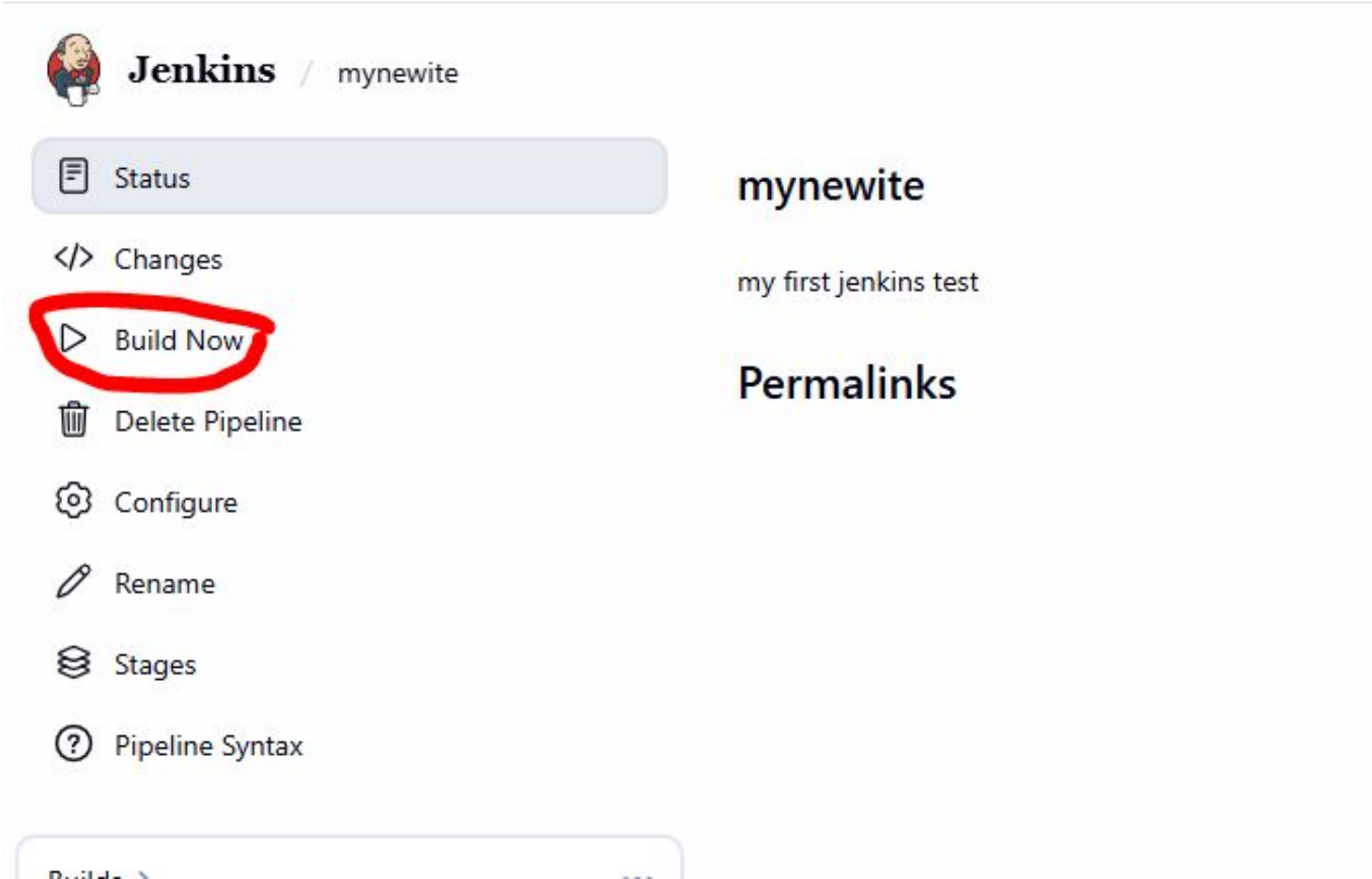
Script ?

```
15     }
16     stage('Test') {
17         steps {
18             // Example of executing tests
19             sh 'mvn test'
20         }
21     }
22     stage('Deploy') {
23         steps {
24             echo 'Deploying application to production...'
25         }
26     }
27 }
```

Save

Apply

Save and click build



Build output

The screenshot displays the Jenkins web interface for a job named "mynewite". The browser address bar shows "localhost:8080/job/mynewite/". The Jenkins logo and job name "Jenkins / mynewite" are at the top. A sidebar on the left contains several actions: "Status" (highlighted), "Changes", "Build Now", "Delete Pipeline", "Configure", "Rename", "Stages", and "Pipeline Syntax". On the right, the job name "mynewite" is displayed, followed by the build name "my first jenkins test" and a section titled "Permalinks". The main content area shows a "Builds" section, which is circled in red. This section indicates "Today" and lists a successful build "#7" at "9:26 PM".

localhost:8080/job/mynewite/

Jenkins / mynewite

Status

Changes

Build Now

Delete Pipeline

Configure

Rename

Stages

Pipeline Syntax

mynewite

my first jenkins test

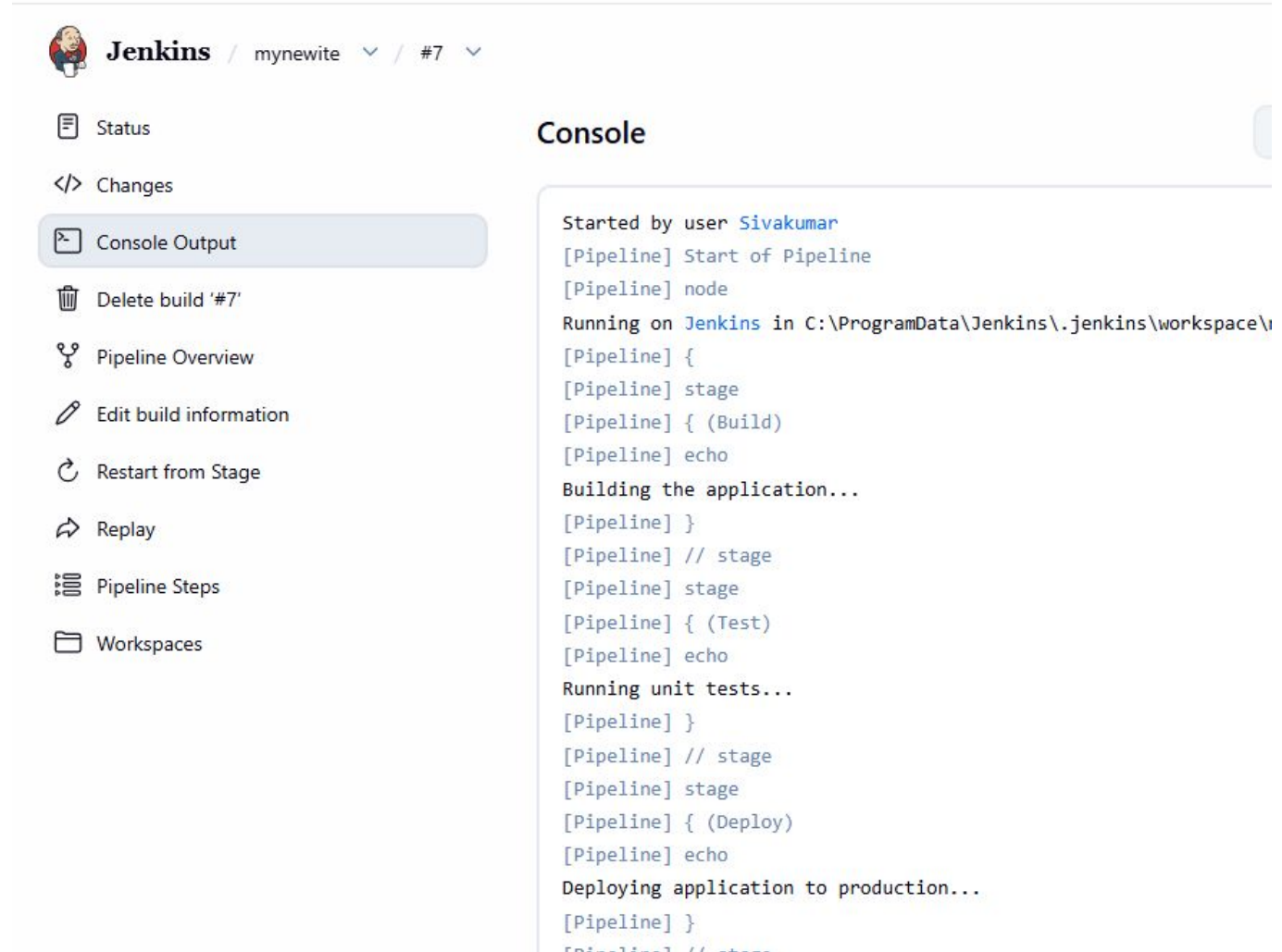
Permalinks

Builds >

Today

✓ #7 9:26 PM

View stages by clicking build output



The image shows the Jenkins web interface for a specific build. At the top, the Jenkins logo is followed by the text "Jenkins / mynewite / #7". Below this is a sidebar with several navigation options: "Status", "Changes", "Console Output" (which is highlighted), "Delete build '#7'", "Pipeline Overview", "Edit build information", "Restart from Stage", "Replay", "Pipeline Steps", and "Workspaces". The main area on the right is titled "Console" and displays the build's output log. The log shows the pipeline starting, running on Jenkins, and executing several stages: "Start of Pipeline", "node", "stage", "(Build)", "echo", "Building the application...", "stage", "(Test)", "echo", "Running unit tests...", "stage", "stage", "(Deploy)", "echo", and "Deploying application to production...".

Jenkins / mynewite / #7

- Status
- Changes
- Console Output**
- Delete build '#7'
- Pipeline Overview
- Edit build information
- Restart from Stage
- Replay
- Pipeline Steps
- Workspaces

Console

```
Started by user Sivakumar
[Pipeline] Start of Pipeline
[Pipeline] node
Running on Jenkins in C:\ProgramData\Jenkins\.jenkins\workspace\
[Pipeline] {
[Pipeline] stage
[Pipeline] { (Build)
[Pipeline] echo
Building the application...
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Test)
[Pipeline] echo
Running unit tests...
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Deploy)
[Pipeline] echo
Deploying application to production...
[Pipeline] }
```

Pipeline overview

← → ↻ ⓘ localhost:8080/job/mynewite/7/console

 **Jenkins** / mynewite ▾ / #7 ▾

-  Status
-  Changes
-  Console Output
-  Delete build '#7'
-  Pipeline Overview
-  Edit build information
-  Restart from Stage
-  Replay
-  Pipeline Steps
-  Workspaces

Console

```
Started by user Sivakumar
[Pipeline] Start of Pipeline
[Pipeline] node
Running on Jenkins in C:\ProgramData\Jenkins\.jenkins
[Pipeline] {
[Pipeline] stage
[Pipeline] { (Build)
[Pipeline] echo
Building the application...
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Test)
[Pipeline] echo
Running unit tests...
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Deploy)
[Pipeline] echo
Deploying application to production...
[Pipeline] }
```



Jenkins

/ mynewite / #7 / Stages



#7



Rerun



Started by Sivakumar Started 2 min 39 sec ago Took 0.85 sec

Start



Build



Test



Deploy



Post Actions

End



Search



Post Actions



81ms



Started 2m 38s ago



Jenkins



Pipeline execution complete. >

32ms



The build was successful! v

27ms



0 The build was successful!



Build 46ms



Test 27ms



Deploy 35ms



Post Actions 81ms

Step 4: Create a Git Repository

- Store your application code in GitHub, GitLab, or Bitbucket.
- my-app/
 - |
 - |— src/
 - |— pom.xml
 - |— Dockerfile
 - |— Jenkinsfile

Step 5: Connect Jenkins with GitHub

1. Create a new Pipeline job.
2. Enter the Git repository URL.
3. Add Git credentials if the repository is private.
4. Save the configuration.

Step 6: Create a Jenkins Pipeline

- Add a file named:
 - Jenkinsfile

```
pipeline {
  agent any

  stages {
    stage('Checkout') {
      steps {
        git 'https://github.com/username/my-app.git'
      }
    }

    stage('Build') {
      steps {
        sh 'mvn clean package'
      }
    }

    stage('Test') {
      steps {
        sh 'mvn test'
      }
    }

    stage('Deploy') {
      steps {
        echo 'Deploying application...'
      }
    }
  }
}
```

Step 7: Configure Build Trigger

- Go to:
- **Job → Configure → Build Triggers**
- Enable:
- GitHub webhook trigger
- Poll SCM (optional)
- Example:
- H/5 * * * *
- This checks for changes every 5 minutes.

Step 8: Configure GitHub Webhook

- In GitHub:
 - Repository
 - ↓
 - Settings
 - ↓
 - Webhooks

- Add webhook:
 - <http://your-jenkins-server/github-webhook/>
 - application/json
 - Choose:
 - Just the push event
 - Save the webhook.

- **Step 9: Build the Application**

- When code is pushed:

- Developer

- |

- ▼

- GitHub

- |

- ▼

- Webhook

- |

- ▼

- Jenkins

- |

- ▼

- Build

Jenkins executes:
mvn clean package